Hidetaka KOBAYASHI, PhD

Assistant Professor (project)
Faculty of Science, University of Toyama
A417, School of Science Bldg., 3190
Gofuku,Toyama, Toyama, Japan 930-8555
Email: hidekoba<at>sci.u-toyama.ac.jp
http://www3.u-toyama.ac.jp/hidekoba/
(updated: Oct 25, 2026)

Personal data:

Born: 10 May 1989, Japan
My name in Chinese characters: "小林 英貴"

Education:

− Ph.D., Earth and Planetary Science, the University of Tokyo, Japan, Mar. 2018.

Title: "Role of ocean carbon cycle in glacial reduction of atmospheric carbon dioxide concentration"
(Advisor: Akira Oka)

− M. Sc., Earth and Planetary Science, the University of Tokyo, Japan, Mar. 2014.
− B. Sc., Department of Earth and Planetary Physics, the University of Tokyo, Japan, Mar. 2012.

Research Experiences:

− Apr. 2018 – Jul. 2022: Postdoctoral Project Researcher.
Affiliation: Atmosphere and Ocean Research Institute, University of Tokyo
Main subject: Ocean biogeochemical cycle modeling
− Aug. 2022 − present: Assistant Professor (Project).
Affiliation: Faculty of Science, University of Toyama
Main subject: Ocean biogeochemical cycle modeling

Publications:

Refereed Scientific Journals:

− **Kobayashi, H.**, T. Yokohata, A. Yamamoto, T. Hajima, H. Tatebe, A. Oka, and K. B. Rodgers, Multi-centennial ocean biogeochemical responses to extended Shared Socioeconomic Pathways, *Biogeosciences*. https://doi.org/10.5194/egusphere-2026-120, 2026.
− Nishida, M., **H. Kobayashi**, and A. Oka, Impact of ocean physical conditions on oceanic carbon pumps and atmospheric carbon dioxide at the Last Glacial Maximum, *Paleoceanography and Paleoclimatology*, **41**, e2026PA005487. https://doi.org/10.1029/2026PA005487, 2026.
− **Kobayashi, H.**, A. Oka, T. Obase, A. Abe-Ouchi, Assessing transient changes in the ocean carbon cycle during the last deglaciation through carbon isotope modeling, *Climate of the Past*, **20**, 769–787,

doi:10.5194/cp-20-769-2024, 2024.
− **小林英貴**，岡顕，氷期の海洋炭素循環シミュレーション，*地球化学*, **57**(2), 205–223, doi:10.14934/chikyukagaku.57.205, 2023.
− Sasaki, Y., **H. Kobayashi**, A. Oka, Global simulation of dissolved 231Pa and 230Th in the ocean and the sedimentary 231Pa/230Th ratios with the ocean general circulation model COCO ver4.0, *Geoscientific Model Development*, **15**, 2013–2033, doi:10.5194/gmd-15-2013-2022, 2022.
− Lhardy, F. et al. [**H. Kobayashi** listed in 11th author], A first intercomparison of the simulated LGM carbon results within PMIP-carbon: role of the ocean boundary conditions, *Paleoceanography and Paleoclimatology*, **36**, e2021PA004302, doi:10.1029/2021PA004302, 2021.
− **Kobayashi, H.**, A. Oka, A. Yamamoto, A. Abe-Ouchi, Glacial carbon cycle changes by Southern Ocean processes with sedimentary amplification, *Science Advances*, **7**(35), eabg7723, doi:10.1126/sciadv.abg7723, 2021.
− **Kobayashi, H.**, and A. Oka, Response of atmospheric pCO_2 to glacial changes in the Southern Ocean amplified by carbonate compensation, *Paleoceanography and Paleoclimatology*, **33**(11), 1206–1229, doi:10.1029/2018PA003360, 2018.
− **Kobayashi, H.**, A. Abe-Ouchi, and A. Oka, Role of Southern Ocean stratification in glacial atmospheric CO_2 reduction evaluated by a three-dimensional ocean general circulation model, *Paleoceanography*, **30**(9), 1202–1216, doi:10.1002/2015PA002786, 2015.

Awards and Fellowships:

− Student Outstanding Presentation Award in the Atmospheric and Hydrospheric Sciences Section, Japan Geoscience Union meeting, 2016.
− Student Outstanding Presentation Award in the Fall meeting, The Oceanographic Society of Japan, 2015
− Student Outstanding Presentation Award in the Atmospheric and Hydrospheric Sciences Section, Japan Geoscience Union meeting, 2015.

Invited Presentations:

− Session: Advanced understanding of Quaternary and Anthropocene hydroclimate changes in East Asia, Annual meeting of Geochemical Society of Japan, FY 2024.
− Polar Research Symposium: “Climate Change and Ecosystems at the Poles during the Neogene”, FY 2022.
− Joint Research Program at the Atmosphere and Ocean Research Institute: Integration of Proxies and Models in Paleoclimate Research: Climate Change During Warm Periods, FY 2022.
− The 69th Annual Meeting of the Geochemical Society of Japan, FY 2022.
− The 8th Annual Meeting of the Paleosciences Society of Japan, FY 2022.
− Session: Ocean geochemistry, Annual meeting of Geochemical Society of Japan, FY 2022.
− Session: Ice cores and paleoenvironmental modeling, Japan Geoscience Union meeting, 2022.
− Session: Paleoclimatology and paleoceanography, Japan Geoscience Union meeting, FY 2018.
− Session: Global climate change driven by the Southern Ocean and the Antarctic Ice Sheet, Japan Geoscience Union meeting, FY 2016.

Presentations at international conferences:

− **Kobayashi, H.**, Role of riverine carbon in the ocean carbon cycle evaluated by an ocean general circulation model with a sedimentary module, May 27, 2026, Makuhari, JpGU-AGU Joint Meeting 2026, poster.

− **Kobayashi, H.**, Atmospheric Δ14C and pCO2 controls on ocean radiocarbon distributions during the Last Glacial Maximum and the last deglaciation, May 26, 2026, Makuhari, JpGU-AGU Joint Meeting 2026, poster.

− **Kobayashi, H.**, T. Obase, A. Oka, and A. Abe-Ouchi, Contrasting changes in atmospheric pCO2 trajectories due to different temporal variations in the Atlantic meridional overturning circulation during the past two deglaciations, May 19, 2026, Bremen, PalMod IOSC 2026, poster.

− **Kobayashi, H.**, S. Sherriff-Tadano, T. Obase, A. Oka, A. Abe-Ouchi, P. Testorf, and F. Pöppelmeier, Background-state dependence of Atlantic Meridional Overturning Circulation, carbon isotopes, and $^{231}Pa/^{230}Th$ under North Atlantic freshwater forcing, May 11, 2026, Bern, PAGES IMPACT workshop 2026, poster.

− **Kobayashi, H.**, A. Oka, T. Obase, M. Nishida, and A. Abe-Ouchi, Diagnosing deglacial ocean carbon cycle change through radiocarbon and stable carbon isotopes, May 7, 2026, Vienna, EGU General Assembly 2026, poster.

− **Kobayashi, H.**, T. Obase, A. Oka, and A. Abe-Ouchi, Contrasting AMOC–carbon cycle linkages across the last two deglaciations, Mar 11, 2026, hybrid (Kyoto), CMIP Community Workshop 2026, poster.

− **Kobayashi, H.**, T. Yokohata, A. Yamamoto, T. Hajima, H. Tatebe. A. Oka, and K. Rodgers, Multi-centennial evolution of ocean biogeochemistry under extended SSP simulations, Mar 5, 2026, Tokyo, TIPMIP General Assembly 2026, poster.

− **Kobayashi, H.**, T. Yokohata, A. Yamamoto, T. Hajima, H. Tatebe. A. Oka, and K. Rodgers, Multi-Centennial Ocean Biogeochemical Responses to Extended Shared Socioeconomic Pathways, Feb 26, 2026, Glasgow, Ocean Sciences Meeting 2026, poster.

− **Kobayashi, H.**, Modeling the ocean carbon cycle across the Last Glacial Maximum and deglaciation: the role of Southern Ocean stratification and carbonate sedimentation, Sept 26, 2025, virtual, PMIP WINGS, oral.

− **Kobayashi, H.**, A. Oka, T. Obase, and A. Abe-Ouchi, Different responses of the ocean carbon cycle during two past interglacials and glacial terminations, Sept 1, 2025, Bengaluru, 15th International Conference on Paleoceanography, poster.

− **Kobayashi, H.**, A. Oka, T. Obase, and A. Abe-Ouchi, Marine carbon cycle responses across interglacial periods: Insights from past warm climates, Jul 24, 2025, Busan, Busan IAMAS-IACS-IAPSO Joint Assembly, oral.

− **Kobayashi, H.**, A. Oka, T. Obase, and A. Abe-Ouchi, Ocean carbon cycle dynamics during the last glacial period: comparative analysis of the MIROC4m model and proxies, Dec 13, 2024, Wasington, D.C., 2024 AGU Fall Meeting, poster.

− **Kobayashi, H.**, Challenging the marine-centric view of the Late Pleistocene carbon cycle, Aug 27, 2024, Ginowan, EPPC Workshop 2024 ~Insight from Snowball into paleo and modern~, oral.

− **Kobayashi, H.**, A. Oka, T. Obase, and A. Abe-Ouchi, Carbon isotope modeling implications for changes in the Southern Ocean carbon cycle during the last deglaciation, Aug 23, 2024, Chicago, 2024 Goldschmidt Conference, oral.

− **Kobayashi, H.**, A. Oka, T. Obase, A. Abe-Ouchi, Assessing transient changes in the ocean carbon cycle during the last deglaciation through carbon isotope modeling, Apr 19, 2024, Vienna, EGU General Assembly, oral.

− **Kobayashi, H.**, A. Oka, T. Obase, A. Abe-Ouchi, Transient response of the ocean carbon cycle during the last deglaciation ~ Investigating the impact of climate and AMOC on carbon isotope signatures ~, July 2023, Rome, XXI INQUA Congress, poster.

− **Kobayashi, H.**, A. Oka, A. Yamamoto, A. Abe-Ouchi, Glacial carbon cycle changes by Southern Ocean processes with sedimentary amplification, July 2023, Berlin, 28th General Assembly of the IUGG, oral.

− **Kobayashi, H.**, A. Oka, A. Yamamoto, and A. Abe-Ouchi: Response of atmospheric pCO_2 to glacial changes in the Southern Ocean amplified by carbonate compensation, 14th International Conference on Paleoceanography (ICP14), Bergen, August 2022, poster.

− **Kobayashi, H.**, A. Oka, A. Yamamoto, and A. Abe-Ouchi, Impacts of enhanced salinity stratification and iron fertilization in the Southern Ocean on the distribution of ocean biochemical tracers during the Last Glacial Maximum, Ocean Sciences Meeting 2022, virtual, March 2022

− **Kobayashi, H.**, A. Oka, A. Yamamoto, and A. Abe-Ouchi, Glacial ocean carbon cycle changes caused by Southern Ocean processes with sedimentary amplification, PMIP 2020, Nanjing/virtual, October 2020, poster.

− **Kobayashi, H.**, and A. Oka, A modeling study of early diagenesis using an ocean sediment model, JpGU-AGU Joint Meeting 2020, virtual, July 2020, iposter.

− **Kobayashi, H.**, A. Oka, A. Yamamoto, and A. Abe-Ouchi, Glacial ocean carbon cycle changes caused by enhanced stratification in the Southern Ocean and iron fertilization from glaciogenic dust, JpGU-AGU Joint Meeting 2020, virtual, July 2020, iposter.

− **Kobayashi, H.**, and A. Oka, Response of atmospheric pCO_2 to glacial changes in the Southern Ocean amplified by carbonate compensation, Ocean Sciences Meeting, San Diego, February 2020, oral.

− **Kobayashi, H.** and A. Oka: Response of atmospheric pCO_2 to glacial changes in the Southern Ocean amplified by carbonate compensation, 13th International Conference on Paleoceanography (ICP13), Sydney, September 2019, poster.

− **Kobayashi, H.** and A. Oka: Response of atmospheric pCO_2 to glacial changes in the Southern Ocean amplified by carbonate compensation, EGU General Assembly, Vienna, April 2019, poster.

− **Kobayashi, H.** and A. Oka: Simulations of glacial ocean carbon cycle with a parameterization of brine rejection, 1st GRAntarctic International Symposium, Tachikawa, December 2018, oral

− **Kobayashi, H.**: Role of carbon cycle in glacial reduction of atmospheric carbon dioxide concentration, Ocean circulation and carbon cycling (OC3) workshop, Cambridge, September 2018, poster.

− **Kobayashi, H.** and A. Oka: Modeling of glacial ocean carbon cycle by focusing on the role of the Southern Ocean and carbonate compensation process, 12th International Conference on Paleoceanography (ICP13),

Utrecht, August 2016, poster.

− **Kobayashi, H.** and A. Oka: Role of Southern Ocean in glacial atmospheric CO_2 reduction, Goldschmidt 2016, Yokohama, June 2016, oral.

− **Kobayashi, H.** and A. Oka: Role of Southern Ocean stratification in glacial atmospheric CO_2 reduction, 26th General Assembly of the IUGG, Prague, June 2015, oral.

− **Kobayashi, H.** and A. Oka: Role of Southern Ocean stratification in glacial atmospheric CO_2 reduction, 2014 AGU Fall Meeting, San Francisco, December 2014, poster.

Funding Information:

Current:

− 2025–2026: JSPS Kakenhi Grant-in-Aid for Transformative Research Areas (A), PI (Area Organizer: Hiroyasu Hasumi), Area title: “Macro coastal oceanography: integrated simulation for the material dynamics from the land through the open ocean”, Title: “Quantifying the Role of Riverine Nutrient and Carbon Transport on the Oceanic Carbon Budget”, 3.9 million yen

− 2024–2028: JSPS Kakenhi Grant-in-Aid for Transformative Research Areas (A), Co-PI (Area Organizer: Shigeru Aoki, PI: Akira Oka), Area Title: “Global Antarctic Science: connecting the chain of changing huge ice sheet and global environments”, Title: “Modeling of climate system dynamics in the Antarctic”

− 2023–2026: JSPS Kakenhi, Grant-in-Aid for Scientific Research (B), Co-PI (PI: Jing Zhang), Title: “The Transport Processes of Nutrients Supplied to the Euphotic Zone via Turbulent Mixing in the Kuroshio Current Region and Their Contribution to Bioproductivity”

− 2023–2026: JST Strategic Basic Research Programs, Research Proposals for the new Research Area, PI (Area Organizer: Jota Kanda), Area title: “Exploration of the Blue Biosphere-Climate System Towards Carbon Cycle Management”, Title: “Elucidating the mechanisms of transient changes in the millennial-scale ocean carbon cycle”, 28.1 million yen

Ended:

− 2023–2025: EMECS Grants-in-Aid for young researchers, PI, Title: “Understanding Carbon and Nutrient Dynamics in Coastal Areas, with a Focus on Carbon Budget Analysis: A Case Study of Toyama Bay”, 3.988 million yen

− 2021–2023: JSPS Kakenhi, Grant-in-Aid for Young Scientists, PI, Title: “Study on the Mechanisms of Glacial-Period Variability in the Ocean Carbon Cycle”, 3.1 million yen

Professional Memberships:

− Japan Geoscience Union

− The Oceanographic Society of Japan.

− Paleosciences Society (PALEO10), Japan

− The Japanese Coral Reef Society

− American Geoscience Union

− European Geoscience Union

Journal referee:

− Climate of the Past

− Communications Earth and Environment

− Geoscientific Model Development

− Geophysical Reserach Letters

− Global and Planetary Change

− Nature Communications

− Nature Geoscience

− Paleoceanography and Paleoclimatology

− Progress in Earth and Planetary Science

− Scientific Reports